Design and Simulation of a 20kW NPC inverter for a 1500V BESS in the USA market

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***Abstract*—This report analyzes the design, modeling, simulation and performance evaluation of a three-phase, three-level Neutral-Point Clamped (NPC) inverter designed for a 20 kW Battery Energy Storage System (BESS) intended for the USA market. The inverter is supplied by a 1500 V DC-link and delivers a 480 V line-to-line AC output. Sinusoidal Pulse Width Modulation (SPWM) with Phase Disposition is adopted as the modulation technique, and Silicon Carbide (SiC) MOSFETs are used as power switches for their high efficiency and thermal performance.**

The system is implemented and simulated in PLECS, where key electrical quantities such as pole voltages, line-to-line voltages, phase voltages, phase currents, and DC-link capacitor voltages are analyzed. A spectral analysis of both voltage and current waveforms is carried out to evaluate the Total Harmonic Distortion (THD). The efficiency is calculated by considering only the conduction, switching and auxiliary losses, as per the project guidelines, and a breakdown of losses among all power switches is provided.

The efficiency curve is evaluated from 10 to 100 percentage of the rated load. Additionally, the 4-point efficiency and European weighted efficiency are presented. The thermal behavior of the converter is verified to ensure that the junction temperature of the SiC MOSFETs remains below 100° C under all operating conditions.

The results confirm that the selected topology and control strategy offer good power quality, acceptable THD levels, and high efficiency across the operating range, making this solution suitable for high-performance BESS inverter applications in compliance with USA standards.

***Index Terms*—Battery Energy Storage System (BESS), Three-Level Inverter, Neutral Point Clamped (NPC), Silicon Carbide (SiC) MOSFET, Sinusoidal PWM, Power Electronics, PLECS Simulation, Total Harmonic Distortion (THD), Efficiency Analysis, Grid-Connected Inverter.**

I. Introduction

Battery Energy Storage Systems (BESS) are becoming a key component in modern electrical grids, especially for applications requiring peak shaving, load leveling, and grid support. A critical component in these systems is the power inverter, which is responsible for interfacing DC storage with the AC grid. This report focuses on the design and simulation of a 20kW three-phase inverter suitable for grid-connected BESS applications in the USA, where the grid standard is 480V AC line-to-line.

The selected topology is a three-level Neutral Point Clamped inverter (NPC), which offers significant advantages over the conventional two-level inverter, including reduced voltage stress on the switches, improved waveform quality, and lower Total Harmonic Distortion (THD). These features make NPC inverters particularly well-suited for high-power applications such as BESS, where energy efficiency and grid compatibility are critical.

Compared to T-type or fFying Capacitor inverters, the NPC structure offers a good compromise between complexity and performance. However, its implementation requires careful balancing of the neutral point and adds additional components such as clamping diodes.

This report explores the modeling of the NPC inverter using PLECS, focusing on key metrics such as voltage and current waveforms, THD, switch losses, and system efficiency. The use of Silicon Carbide (SiC) MOSFETs, known for their superior switching performance and thermal properties, further enhances the overall efficiency and compactness of the converter.

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II. Converter Topology and Modulation Technique

An inverter is a power electronic device used to provide a variable AC output voltage from a constant DC source, commonly a three-phase AC voltage. The inverter in which the number of available levels n is greater than 2 are called 'Multilevel inverters'. They allow increasing power ratings, improving the output voltage in terms of waveforms, harmonic content, and THD since they can better approximate the reference signal, reducing electromagnetic interference emissions, thanks to the fact that the dv/dt is reduced due to the higher number of levels. Compared to the traditional 2-level inverter, the 3-level inverter has the half of the voltage stress on the switching components for the same maximum output voltage and generates lower harmonics for the same frequency. The basic topologies of the Multilevel Voltage Source inverter are:

- Neutral Point Clamped (NPC)
- Flying Capacitor (FC)
- Cascaded H-Bridge (CHB)

A. The Neutral Point Clamped (NPC) inverter

The power stage implemented in this project is a three-phase, three-level Neutral Point Clamped (NPC) inverter. Compared to the conventional two-level topology, the NPC structure offers reduced voltage across each switch and lower harmonic distortion in the output voltage, making it suitable for high voltage applications such as a 20 kW Battery Energy Storage System (BESS). Each phase leg of the NPC consists of four power switches and two clamping diodes, which allow the output to reach three voltage levels: $+V_{dc}/2$, 0 and $-V_{dc}/2$. This enables finer control of the output waveform

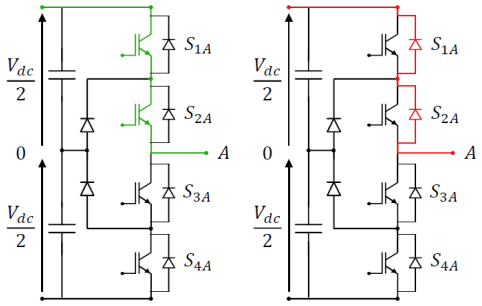


Fig. 1. Positive operating stage

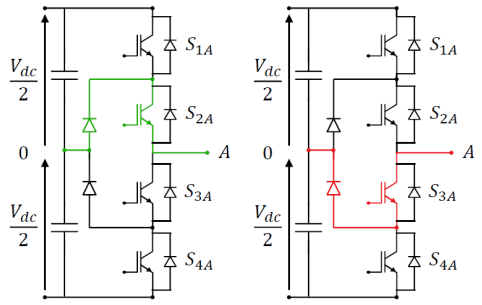


Fig. 2. Zero operating stage

and reduces dv/dt stress on the load and filter components. In this project, Silicon Carbide (SiC) MOSFETs (Wolfspeed C3M0021120K) were selected due to their high switching frequency capabilities, low losses, and suitability for 1500 V DC-link systems. Usually, instead of having two different DC sources on the input side, it is common to have two capacitors and only one DC source. The drawback of having two capacitors is related to the presence of some oscillations on the 0 point: these oscillations are due to the currents that are present only when we have a pole voltage equal to 0, since when the pole voltage is equal to $+V_{dc}/2$ or $-V_{dc}/2$ the current doesn't flow through the capacitors. The bigger the capacitors, the smaller the oscillations. The two capacitors are self-balanced: this means that even if their initial condition is not perfectly equal to $+V_{dc}/2$ or $-V_{dv}/2$, they tend to reach that specific value. It's only a matter of time. This is a greater advantage since having a capacitor voltage that diverges could be s dangerous problem for the component life.

The operating stages of a single leg of the NPC inverter are:

1) $V_{out} = +V_{DC}/2$: switches Q_1 and Q_2 are turned ON, while Q_3 and Q_4 remain OFF, as in Figure 1.
   - For positive load current ($i_A > 0$), current flows through Q_1 and Q_2.
   - For negative current ($i_A < 0$), freewheeling diodes conduct.
2) $V_{out} = 0$: Current flows through the clamping diodes, as in Figure 2.
   - For $i_A > 0$, the conduction path includes D_1.
   - For $i_A < 0$, the conduction path includes D_2.
3) $V_{out} = -V_{DC}/2$: switches Q_3 and Q_4 are ON, while Q_1 and Q_2 are OFF, as in Figure 3.

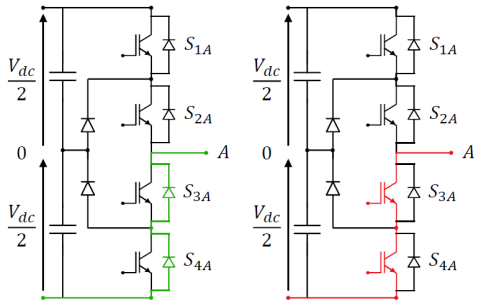


Fig. 3. Negative operating stage

   - For $i_A > 0$, current flows through the freewheeling diodes.
   - For $i_A < 0$, current flows directly through Q_3 and Q_4.

These operational modes validate the NPC's functionality as an inverter and also highlight one of its main drawbacks: uneven stress distribution among switches. Switches Q_2 and Q_3 operate more frequently than Q_1 and Q_4, leading to thermal unbalance, accelerated aging, and increased conduction losses over time. Proper thermal management and switching balance strategies are therefore crucial to ensure long-term reliability.

B. Sinusoidal Pulse Width Modulation - Phase Disposition (SPWM-PD)

The inverter is driven by using Sinusoidal Pulse Width Modulation (SPWM) with Phase Disposition (PD), a method commonly used in three-level topologies. Although all Level-Shift (LS) PWM techniques could be applied, only the PD technique ensures maximum voltage levels. In SPWM-PD, three sinusoidal reference signals (one for each phase) are compared with two high-frequency triangular carriers that are aligned in phase (i.e., both carriers have the same phase but different amplitude ranges). The carrier signals divide the modulation region into three voltage levels. As in Figure 4, the output voltage of each phase is determined based on the result of this comparison:

- if the reference signal is greater than the upper carrier: $+V_{dc}/2$
- if it is lower than the lower carrier: $-V_{dc}/2$
- otherwise: 0

This modulation scheme ensures minimal common-mode voltage and good harmonic performance.
A modulation index of 0.52 is chosen to match the desired line-to-line RMS voltage (480 V) for 1500V DC-link system. The carrier frequency is set to 16 kHz and the reference frequency is 60 Hz, matching the USA grid. This modulation strategy results in balanced three-phase output voltages with reduced Total Harmonic Distortion (THD), and it is compatible with standard passive filtering to further smooth the output.

III. Device Selection

The design process starts by identifying and analyzing the main constraints which are summarized in Table I.

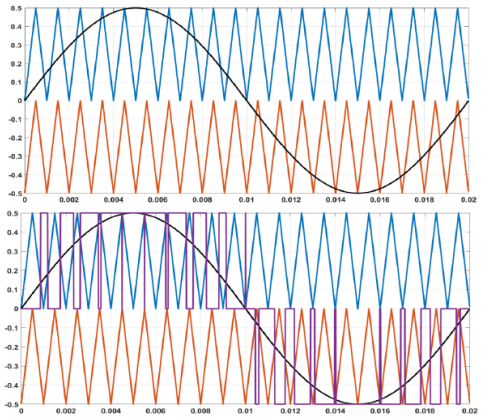


Fig. 4. Single leg PD

TABLE I
CONSTRAINTS

Rated DC Voltage	1500 V
Rated Power	20 kW
Line-to-line Voltage	480 V
Frequency	60 Hz
Technology	SiC-MOSFET
Max Junction Temperature	100°C

In an NPC topology inverter, each power switch is required to block only half of the total DC-link voltage, $V_{DC}/2$. The required V_{DS} rating should be at least 750V. The second parameter is the DC drain current I_D which is defined as the continous drain-source current that causes the MOSFET junction temperature to reach its maximum allowable value, starting from a specified temperature. Even if the SiC-MOSFET junction temperature is greater compared to other power switches, the maximum junction temperature specified is quite low, so a conservative approach is adopted. It was assumed that each device conducts for the entire switching period, and therefore, the required I_D is approximated by the RMS value of the phase current at rated power. Assuming a unity power factor, the line current is calculated as:

$$I_L = \frac{P_{\text{rated}}}{\sqrt{3} \cdot V_{LL}} \approx 24A \tag{1}$$

Consequently, a device with a rated I_D = 30A was selected to ensure thermal reliability under continous conduction.
The choice is the SiC MOSFET of the Wolfspeed family, C3M0021120K combined with a fast-recovery anti-parallel diode. The main parameters of these SiC-MOSFET are shown in Figure II.
The following device selection regarded the clamping diodes, in order to minimize the shipment costs, a suitable components from the same productor can be investigated, and for the same considerations on the blocking voltage, the forwars current and the maximum junction temperature a conservative approach was considered selecting the C4D20120D which parameters are listed in Table III. In the Table IV the junction to case thermal resistances $R_{th,jc}$ of these devices are collected for the different devices. In the Table V are listed the unit prices

TABLE II
WOLFSPEED C3M0021120K SIC-MOSFET

Drain-Source Voltage Vds	1200V
Continous Id	74A
Rds(on)	0.021
Max junction temperature	175°C

TABLE III
WOLFSPEED C4D20120D SHOTTKY DIODES

DC blocking voltage	1200V
Forward Current	34A
Forward Voltage	1.5V
Maximum Junction Temperature	175°C

of each device considering as order volume the number of devices that would be needed to build one inverter.

IV. NUMERICAL VALIDATION

The model of the three-level NPC inverter has been implemented and simulated in PLECS. The simulation results confirm the correct behavior of the modulation strategy and the inverter's ability to deliver the required 20 kW output power to the grid.

First of all the modulating index m has been calculated in order to achieve at the inverter output the required phase voltage V_{ph} by using the formula:

$$m = 2\sqrt{2}\frac{V_{ph}}{V_{DC}} \tag{2}$$

The desired phase voltage is given by the line-to-line voltage divided by the root of 3, which in this case by design constraints is 277 A, so the modulating index m results in:

$$m = 2\sqrt{2}\frac{277}{1500} = 0.52 \tag{3}$$

In order to simulate the grid, a three-phase generator has been connected to the AC-side through coupling inductors L and resistors R.
Considering a single phase, the active power is linked to the displacement factor ϕ between the inverter and the grid phase voltages and their amplitude with the following relation:

$$P = \frac{1}{\sqrt{2}}\frac{V_{DC}V_{ph}}{4L\omega}sin\phi \tag{4}$$

where $\omega = 2\pi f$ is the angular frequency. This phase displacement has been implemented acting on the angles of the

TABLE IV
JUNCTION TO CASE THERMAL RESISTANCE VALUES

SiC-MOSFET	0.32°C/W
Diodes	0.43°C/W

TABLE V
DEVICES UNIT PRICES

C3M0021120K	12	26.11
C4D20120D	18	12.17

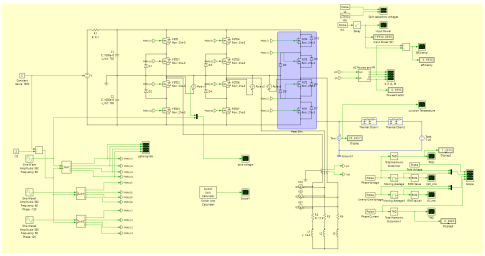


Fig. 5. PLECS overview

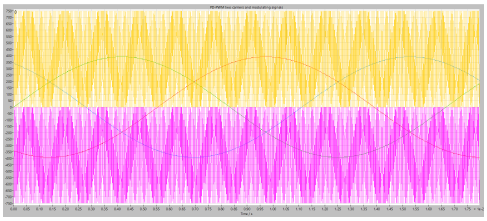


Fig. 6. PD-SPWM Two carriers and modulating signals

modulating signals of the SPWM. The power factor PF has been considered fixed constant to 1. The active, reactive and apparent power were measured at the inverter output by using a three-phase meter based on instantaneous power calculations.

The generic schematics of the NPC inverter is shown in the Figure 5.

The parameters used are the following:

- $V_{dc} = 1500V$
- $f_{sw} = 16kHz$
- $C = 1mF$
- $R = 11.5$
- $L = 1mH$

A. Main Quantities Waveforms

In Figure 6 are shown the two triangular carriers and the modulating signals of each leg. With the logic control described before for the SPWM-PD we obtain the following gate signals in Figure 7, plotted only for the first leg just for simplicity.

The pole voltage of each leg has got 3 levels as expected. It follows the sinusoidal profile of its modulating signal, as in Figure 8. The 3 pole voltages are phase-shifted of 120° one from the other as in Figure 9. Due to the presence of the clamped diodes, the static switches of the inverters are subjected to a voltage equal to the one provided by the capacitors and not to the one provided by the DC source. This

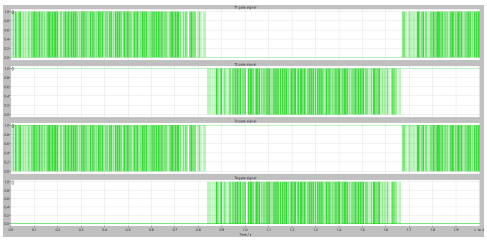


Fig. 7. Gate signals of the first leg

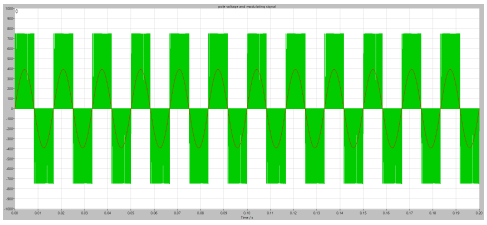


Fig. 8. Pole voltages and modulating signal

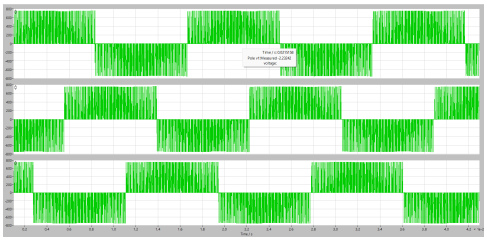


Fig. 9. Pole Voltages

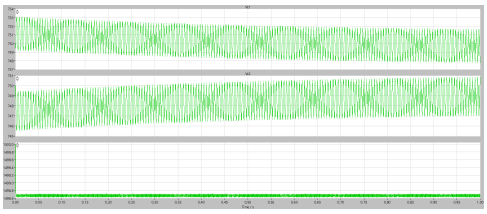


Fig. 10. Self balancing capacitor voltages

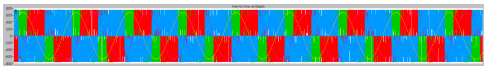


Fig. 11. Line-to-line voltages

is a great advantage, since we can choose smaller components and reduce the commutation losses: in general, if we have n levels, the voltage across the shunts is $V_{dc}/(n-1)$, this means that in the case of the three-level inverter it is equal to $V_{dc}/2$.

The input capacitances are chosen to have a trade-off between a faster self-balancing (low value of C) and less oscillations on the pole voltage (high value). In Figure 10 we see the results we want to obtain, the V_{C1} becomes almost equal to the V_{C2}.

In Figure 11 is plotted the line-to-line voltages.

In Figure 12 we can see the phase voltages and the related Fast Fourier Transform (FFT) shown in Figure 13 , from which we can see that the value at frequency 60Hz is about 392Vp which corresponds to 277Vrms with very low harmonic pollution. In Figure 15 are shown the filtered main voltages. The introduction of a filter imporoves the THD_v which is reduced to 5.4%, instead of the the THD_v of the line-to-line voltage which is about 120%, Figure 14.

The phase currents as expected are 34Ap, 24.5Arms, with very low harmonic pollution by analyzing the FFT, Figure 17 and the total harmonic distortion is about 3.4% after an initial peak of about 7%, Figure 18, which can be considered a good

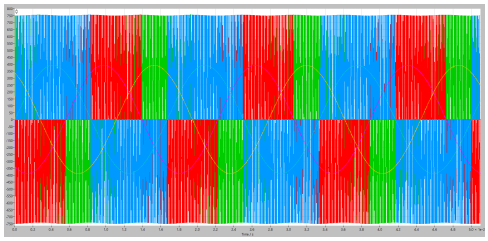


Fig. 12. Phase voltages

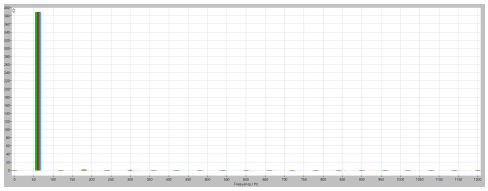


Fig. 13. FFT of phase voltages

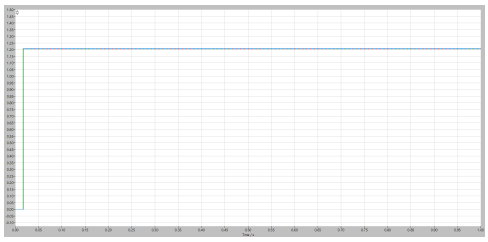


Fig. 14. THD of phase voltages

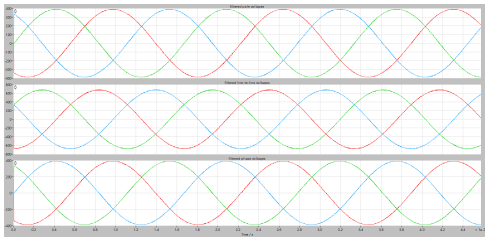


Fig. 15. Filtered voltages

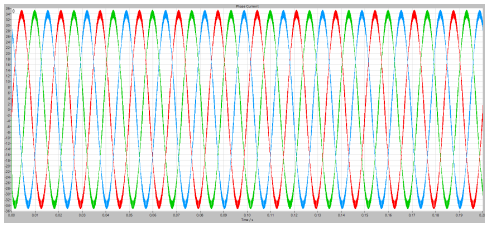


Fig. 16. Phase currents

result. It is also possible to decrease this THD_i by increasing the switching frequency, but this will lead to an increase of the switching losses which are proportionally dependent to the f_{sw}. Moreover, thanks to the 3-phase meter, we can compute the power factor (PF) which is 0.9995 at rated power.

B. Thermal Analysis and Efficiency Calculation

The thermal analysis has been done on a single leg. This has been done as in Figure 19 by introducing a generic small

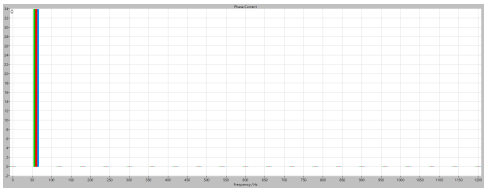


Fig. 17. FFT of phase currents

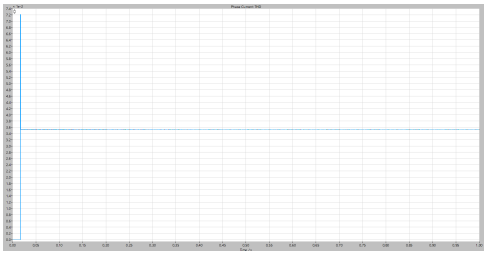


Fig. 18. THD of phase currents

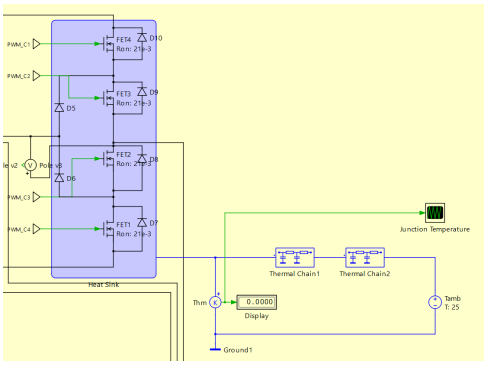


Fig. 19. Temperature simulation

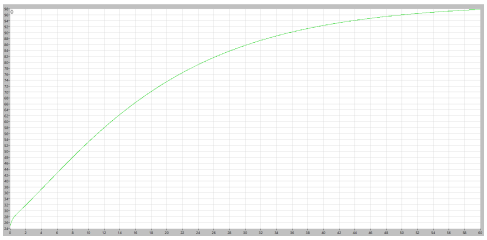


Fig. 20. Junction temperature

heatsink with temperature ambient of 25°C and we can notice that in a time of a minute the temperature measured on the junction remains always under the maximum value of 100°C given as project's constraint and it stabilizes at a values below 98°C, Figure 20.

A comparison between the switching and conduction losses among the various devices of a single leg at rated power is presented in Figure 21. Conduction losses represent the dominant component, which is a notable drawback of the three-level NPC topology, as it inherently requires two devices to conduct simultaneously during each conduction path. Additionally, the two inner switches approximately doubles the losses compared to the outer switches which compromises the reliability of the converter due to the uneven thermal stress between the

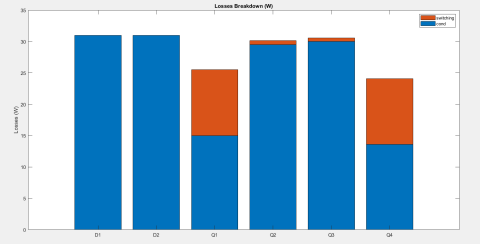


Fig. 21. Losses breakdown

TABLE VI
EFFICIENCY AT DIFFERENT POWER OUTPUT

Power [W]	Efficiency [%]
20000	99.36
15000	99.78
10000	99.72
6000	99.02
5000	98.51
4000	97.66
2000	94.14
1000	92.25

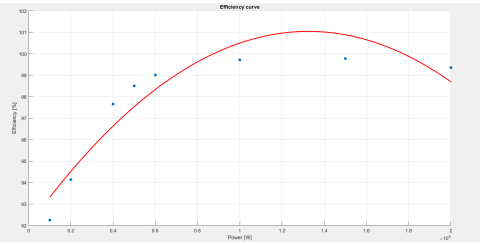


Fig. 22. Inverter efficiency curve

components.

The converter efficiency has been estimated, according to specification, by multiple simulations at different output power. After that we estimated the efficiency in each point using the formula:

$$\eta = \frac{P_o}{P_o + P_{\text{add}} + P_{\text{sw}} + P_{\text{cond}}} \tag{5}$$

The inverse of the efficiency values obtained were used to plot the efficiency from $1000W$ to $20000W$, Table VI a second order polynomial fitting was used in order to estimate the coefficients k_0, k_1 and k_2 in the equation:

$$\eta = \frac{1}{P_o + k_0/P_o + k_1 P_o + k_2 P_o^2} \tag{6}$$

Then a comparison between the obtained efficiency curve and the values used for the fitting is presented in the Figure 22. The polynomial function $\eta(P_o)$ was then used to calculate some standard for efficiency evaluation, the 4-point efficiency is calculated as $\eta_{4-p} = 0.25\eta_{25\%} + 0.25\eta_{50\%} + 0.25\eta_{75\%} + 0.25\eta_{100\%} = 99.34\%$
The European Union efficiency is calculated as $\eta_{EU} = 0.03\eta_{5\%} + 0.06\eta_{10\%} + 0.13\eta_{20\%} + 0.1\eta_{30\%} + 0.48\eta_{50\%} + 0.2\eta_{100\%} = 98.82\%$

V. CONCLUSION

This report presented the modeling, simulation, and performance evaluation of a 20kW, three-phase, three-level Neutral Point Clamped (NPC) inverter designed for a 1500V Battery Energy Storage System (BESS), targeting grid-connected applications in the United States.

The selected NPC topology, driven by Sinusoidal Pulse Width Modulation (SPWM) with Phase Disposition (PD), demonstrated excellent output waveform quality, with reduced Total Harmonic Distortion (THD) and balanced phase voltages. The use of Silicon Carbide (SiC) MOSFETs enabled high switching frequency operation with significantly lower switching and conduction losses compared to traditional silicon devices, while maintaining acceptable thermal limits under nominal load.

Simulation results obtained in PLECS validated the steady-state behavior of the inverter, confirming the proper operation of key functionalities such as DC-link voltage splitting, capacitor self-balancing, and low harmonic current injection. The inverter showed high efficiency across the entire load range, with peak efficiency exceeding 99% and European weighted efficiency above 98%, in line with grid-tied inverter requirements in the USA market.

Overall, the proposed design offers a promising solution for next-generation high-voltage BESS inverters, combining modularity, high efficiency, and grid compliance. Future developments may include the integration filters for reduction of output harmonics, advanced thermal management strategies, and the adoption of hybrid configurations or active neutral point control to enhance system performance and reliability.

VI. ZIP FILE ATTACHMENT

In the .zip attachment the PLECS file, a directory with datasheets and a PDF of the report assignement are present.



Alessandro Gaggioli was born in Pesaro (PU), Italy, in 2000. He received a B.Sc. in Automation Engineering in March 2023 with a mark of 102/110 discussing a thesis titled "Haptic Control in ROS2 ". He is currently enrolled in the M.Sc. in Automation Engineering. His academic interests are related to power electronic converters, electric vehicles, and electric marine propulsion systems.



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